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51 citations · h-index 4

Mingyu Sung

AI Research Engineer — Efficient GenAI Inference (Quantization · Caching · Acceleration)

Nota AI, Platform Team (Quantizer) · South Korea

Research interests	Efficient inference for any model family — LLMs, VLMs, VLAs, diffusion models — via quantization, KV-cache compression, cache-based acceleration, speculative decoding and kernel optimization
Education	Ph.D. in Artificial Intelligence, Kyungpook National University (2026)
Open source	Core contributor to the Nunchaku CUDA acceleration engine (ICLR 2025 Spotlight); 16 merged PRs across Nunchaku, Hugging Face Transformers, ByteDance DreamO and other projects

EXPERIENCE

AI Research Engineer — Platform Team, Quantizer Part 2026.02 – Present

Nota AI

- Develop Nota's LLM/VLM quantization platform: post-training quantization (GPTQ, AWQ, SmoothQuant, rotation-based), KV-cache quantization, and low-bit export (NVFP4, GGUF).
- Extend model support and evaluation across LLM, VLM, MoE, embedding and video-diffusion models.
- Optimized K-EXAONE 236B (MoE) for FuriosaAI's data-center NPU with LG AI Research: ~71% smaller model at ~99.2% accuracy retention.

EDUCATION

Ph.D., Artificial Intelligence 2021.08 – 2026.02

Kyungpook National University (KNU) · Thesis: Fast and Trustworthy Super-Resolution with VLM-Guided Diffusion Model

M.S., Computer Science 2019.09 – 2021.08

Kyungpook National University (KNU) · Thesis: Probabilistic Classification Method of Spiking Neural Network Based on Multi-Labeling of Neurons

B.S., Computer Science 2013.03 – 2019.08

Kyungpook National University (KNU)

OPEN SOURCE

Nunchaku — Core Contributor 2025 – 2026-02

nunchux-ai/nunchaku · 9 merged PRs · CUDA acceleration engine for 4-bit neural networks (ICLR 2025 Spotlight, 3.9k+ stars)

- Cache optimizations — V2 FBCaching (#621), double FB cache, TeaCache batch processing (#601) — for 2–5x additional speedup
- IP-Adapter (XLabs flux-ip-adapter-v2) support (#418)
- Fixes: LoRA key mismatch (#557), ControlNet (#360, #452), Sana (#380), offload segfault (#440)
- Forward-pass tensor-op cleanup (#491)

ComfyUI-nunchaku — Contributor 2025

nunchux-ai/ComfyUI-nunchaku · 2 merged PRs · ComfyUI integration for the Nunchaku engine

- IP-Adapter support for ComfyUI (#305)
- Cache-mechanism file separation (#474)

DreamO — Contributor 2025

bytedance/DreamO · 1 merged PRs · ByteDance's unified image-customization framework

- Dynamic FBCache / DoubleFBCache support for Nunchaku engine integration (#104)

Transformers — Contributor 2026

huggingface/transformers · 3 merged PRs · Hugging Face Transformers library

- Fixed float16 overflow in Gemma4 vision pooler (#46277)
- Fixed Gemma `sliding_window` being halved on every config save/reload, a silent accuracy regression for EmbeddingGemma (#47940)
- Fixed `_init_weights` reading a quantized child's `weight`, which broke loading CLIP-family quantized checkpoints (#47921)

LoopQuant — Contributor 2026

Kishon-zzx/LoopQuant · 1 merged PRs · Post-training quantization for looped language models (Ouro)

- Fixed the LCTC teacher load resetting the ILPA per-loop activation-bit allocation, which ran the W4A6 configs at A4 in every loop; Ouro-2.6B W4A6 WikiText-2 perplexity 8.10 → 7.85 (#1)

PUBLICATIONS

[J1] **A Bi-Objective Framework for Scene Text Super-Resolution: Balancing Legibility and Fidelity**
M. Sung, S. Ham, K. Kim, I.-M. Kim, S. Yun, J.-M. Kang · Transactions on Machine Learning Research (TMLR) · 2026

[J2] **H2-Cache: A Novel Hierarchical Dual-Stage Cache for High-Performance Acceleration of Generative Diffusion Models**
M. Sung, I.-M. Kim, S. Yun, J.-M. Kang · IEEE Open Journal of the Computer Society · 2025

[J3] A Novel VLM-Guided Diffusion Model for Remote Sensing Image Super-Resolution
M. Sung, M.-G. Gong, S.-J. Ham, I.-M. Kim, S. Yun, J.-M. Kang · IEEE Geoscience and Remote Sensing Letters · 2025

[J4] DeCo-MeSC: Deep Compression-Based Memory-Constrained Split Computing Framework for Cooperative Inference of Neural Network
M. Sung, V. Palakonda, I.-M. Kim, S. Yun, J.-M. Kang · IEEE Transactions on Vehicular Technology · 2025

[J5] Generative Diffusion Model-Based Deep Learning Framework for Remaining Useful Life Prediction
S. Ha*, **M. Sung***, F. Saeed, S. Yun, I.-M. Kim, J.-M. Kang · IEEE Internet of Things Journal · 2025

[J6] Entropy-based sampling for efficient training of deep learning on CNC machining dataset
M. Sung, C. Park, S. Ha, M. Ha, H. Lee, J. Kim, J.-M. Kang · Electronics Letters · 2024

[J7] Probabilistic Classification Method of Spiking Neural Network Based on Multi-Labeling of Neurons
M. Sung, J. Kim, J.-M. Kang · Mathematics · 2023

[J8] Training and Inference using Approximate Floating-Point Arithmetic for Energy Efficient Spiking Neural Network Processors
M. Kwak, J. Lee, H. Seo, **M. Sung**, Y. Kim · ICEIC · 2021

[J9] Training spiking neural networks with an adaptive leaky integrate-and-fire neuron
M. Sung, Y. Kim · IEEE ICCE-Asia · 2020

* Equal contribution

PREPRINTS

[P1] RECAP-KV: Single-Prefill Approximation of Reconstruction Importance for Reusable KV Cache Eviction
OpenReview · 2026

[P2] Memory- and Latency-Constrained Inference of Large Language Models via Adaptive Split Computing
M. Sung, V. Palakonda, S. Im, S. Moon, I.-M. Kim, S. Yun, J.-M. Kang · arXiv:2511.04002 · 2025

[P3] No Pose Estimation? No Problem: Pose-Agnostic and Instance-Aware Test-Time Adaptation for Monocular Depth Estimation
M. Sung, H. Choe, I.-M. Kim, S. Yun, J.-M. Kang · arXiv:2511.05055 · 2025

[P4] Why Should the Server Do It All?: A Scalable, Versatile, and Model-Agnostic Framework for Server-Light DNN Inference over Massively Distributed Clients via Training-Free Intermediate Feature Compression
M. Sung, S. Im, D. Bang, I.-M. Kim, S. Yun, J.-M. Kang · arXiv:2511.11608 · 2025

[P5] Range Asymmetric Numeral Systems-Based Lightweight Intermediate Feature Compression for Split Computing of Deep Neural Networks
M. Sung, S. Im, V. Palakonda, J.-M. Kang · arXiv:2511.11664 · 2025

SELECTED PROJECTS

K-EXAONE 236B MoE Optimization (Nota AI × LG AI Research × FuriosaAI) ↗ — Quantization Engineer · 2026
Optimized the 236B-parameter MoE model for FuriosaAI's data-center NPU via targeted precision analysis of degradation-prone sections — ~71% size reduction with ~99.2% accuracy retention (GPQA 79.80 / IFBench 68.98 / AIME25 88.57).

LLM Inference Optimization (Samsung Challenge) — Lead Developer · 2024
Optimized pipeline for Phi-3 using Torch-TensorRT conversion and memory-aware dynamic batching to accelerate LLM inference.

Advanced CNC Machine Tool Diagnosis & Prognosis (KERI) — Lead Developer · 2023
Deep learning models for tool wear classification, anomaly detection, and RUL prognosis.

AI-based Intelligent CCTV (ABB Project) — AI Optimization Engineer · 2023
Optimized an LSTR behavior-detection model for edge devices via Knowledge Distillation (TransKD) and Deep Compression — 25x model compression.

Global Basic Research Laboratory (NRF Project) — Participating Researcher · 2022
Deep learning-based channel estimation (CNN denoising, self-supervised learning) for IRS-aided communication systems.

ML-based Sensor Data Analysis (JS System Project) — Lead Developer · 2022
ML-based sensor feature-importance extraction and model performance optimization.

AWARDS

Best Paper Award · KNU-EE Research Congress — 2025
A Novel VLM-Guided Diffusion Model for Remote Sensing Image Super-Resolution

SKILLS

Research	model quantization, KV-cache compression, diffusion/LLM inference acceleration, split computing, super-resolution, long-context evaluation
Programming	Python, C++, CUDA C, MATLAB
Frameworks & Tools	PyTorch, HuggingFace Transformers, TensorRT, Triton, flash-attn, Docker, Git, LaTeX
Methodology	preregistration, claims ledger, freeze discipline
Languages	Korean (native) · English (business fluent)